



复旦大学生命科学学院

School of Life Sciences Fudan University

课程自用讲义

生命科学科研伦理和规范

2026 春

本讲义采用知识共享署名-非商业性使用-禁止演绎 4.0 国际许可协议 (CC BY-NC-ND 4.0) 发布。您可以自由复制、学习和分发本作品，但必须注明作者姓名，且不得用于商业目的，也不得对本作品进行任何修改或改编。协议全文见：<https://creativecommons.org/licenses/by-nc-nd/4.0/>

《生命科学科研伦理和规范》不为生命科学学院生物科学培养方案的专业核心必修课程，但作为“专业进阶 I”的二选一必修，仍属于必修课。

制作：复旦 23 生科系 若许_lysis

目 录

I	Case Studies	1
1	Case 1	1
2	Case 2	3
3	Case 3	5
4	Case 4	8
5	Case 5	10
6	Case 6	13
7	Case 7	15
8	Case 8	18
9	Case 9	20
10	Case 10	23
11	Case 11	25
12	Case 12	28
II	Key Principles & Personal Reflections	30
13	Key Principles That Demand Special Attention	31

14 Personal Reflections	34
14.1 On Human-Derived Samples and Data	35
14.2 On Data Integrity and Reproducibility	35
14.3 On Collaboration and Authorship	36
14.4 On Dual-Use Concerns and Biosafety	37
14.5 On Publication and Transparency	37
14.6 Final Thought	38

Case Studies

Case 1

Scenario: May, a second-year graduate student, prepares her qualifying exam by copying entire sentences and paragraphs from published papers without using quotation marks. She includes vague citations like "(see ... for more details)" rather than proper attribution. Faculty members notice inconsistencies in writing style, investigate, and confirm the plagiarism. The dean expels May with the option to reapply the following year.

情景：二年级研究生梅在准备资格考试时，直接从已发表的论文中复制整句整段的句子和段落，且未使用引号。她引用原文时也语焉不详，例如“（详见……）”，而非进行规范的标注。教员们注意到她写作风格前后矛盾，展开调查后证实了抄袭行为。院长开除了梅，但允许她次年重新申请。

Analysis

May's behavior constitutes textbook plagiarism: appropriating another person's ideas, processes, results, or words without giving appropriate credit.

梅的行为构成教科书式的剽窃：挪用他人的想法、过程、结果或文字而不给予适当的署名。

The use of "(see ... for more details)" is insufficient attribution —it does not indicate that the actual text was taken verbatim.

使用“（详情请见……）”不足以表明出处——它并不能说明原文是逐字摘录的。

The qualifying exam is meant to assess a student's own understanding and synthesis ability. Copying undermines this educational objective.

资格考试旨在评估学生自身的理解和综合能力。抄袭有悖于这一教育目标。

The inconsistency in writing style is a common red flag that raises suspicion of plagiarism, demonstrating that such acts are often detectable.

写作风格不一致是常见的抄袭警讯，表明这类行为通常是可以被发现的。

The institutional response (expulsion with reapplication option) reflects a balance between severity and rehabilitation.

机构的回应（开除但允许重新申请）体现了严厉惩戒与改过机会之间的平衡。

What I Would Do

If I were in May's position, I would never incorporate verbatim text without quotation marks and proper citations. If I needed to reference published findings, I would paraphrase them in my own words while citing the original sources precisely. If I realized I had made a plagiarism mistake before submission, I would correct it immediately. If I discovered it after the fact, I would proactively report the error to the faculty, acknowledge it openly, and accept the consequences —demonstrating that I understand the severity and am committed to research integrity going forward.

如果我是梅，我绝不会在没有引号和规范引用的情况下使用逐字原文。若需要引用已发表的研究结果，我会用自己的语言改写，并准确标注原始来源。如果我在提交前意识到自己犯了剽窃错误，我会立即更正；如果事后才发现，我会主动向教师报告错误，坦诚承认，并接受相应后果——以此表明我理解问题的严重性，并承诺今后坚守科研诚信。

Bioethics Knowledge Points

Definition of Plagiarism: Using someone else's ideas, processes, results, or words without proper credit —one of the three pillars of research misconduct (Fabrication, Falsification, Plagiarism).

剽窃的定义：使用他人的想法、过程、结果或文字而未给予适当署名——这是科研不端三大类型（编造、篡改、剽窃）之一。

Self-correcting nature of science: The detection of inconsistencies in writing style illus-

trates how the scientific community has self-correcting mechanisms.

科学的自我纠错性：写作风格不一致被发现，说明科学共同体具有自我纠错机制。

Institutional responsibility: Universities must have clear policies and enforcement mechanisms to handle misconduct.

机构责任：大学必须有明确的政策和执行机制来处理不端行为。

Intent vs. negligence: Even if May did not intend to deceive, her careless attribution still constitutes plagiarism. Ignorance of proper citation practices is not a defense.

故意与疏忽：即使梅并非有意欺骗，她草率的引用方式仍构成剽窃。不了解规范引用方法不能作为辩护理由。

Case 2

Scenario: A graduate student faces a difficult relationship with their advisor. The mentor may be unavailable, unresponsive, or have conflicting priorities. The mentee feels stuck — unsure whether to address the problem directly, seek help from the department, or simply endure the situation. The student's career development, emotional well-being, and academic progress are all at stake.

情景：一名研究生与导师关系紧张。导师可能没有时间、回应迟缓，或有相互冲突的优先事项。学生感到进退两难——不确定是直接沟通、向院系求助，还是默默忍受。学生的职业发展、情绪健康和学业进展都受到影响。

Analysis

The mentor-mentee relationship is built on trust, mutual respect, and a shared commitment to growth and development.

导师—学生关系建立在信任、相互尊重以及共同致力于成长和发展的基础之上。

It is primarily the mentee's responsibility to work out the difficult situation with their mentor, as the mentee is most impacted.

解决与导师之间的困难处境，首先是学生本人的责任，因为受影响最大的正是学生。

The mentee is the first responsible person for their own career development, whether academic or industrial.

无论未来走学术道路还是产业道路，学生都是自己职业发展的第一责任人。

Department, school, and university all bear responsibility for monitoring the supervisor-trainee relationship.

院系、学院和大学都负有监督导师—受训者关系的责任。

Extreme or impulsive actions (e.g., quitting without a plan, public confrontation) are unwise and may harm one's own career.

极端或冲动的做法（例如没有计划就退学、公开对抗）并不明智，可能损害自己的职业前景。

What I Would Do

I would first attempt direct, respectful communication with my mentor —scheduling a formal meeting to discuss my concerns and propose constructive solutions. I would prepare a written research progress report to give the conversation a professional framework. If direct communication failed, I would seek advice from a thesis committee member, a departmental ombudsman, or the graduate school's student affairs office. Throughout, I would keep documented records of my communications and decisions. I would remind myself: "There are always more ways out than problems."

我会首先尝试与导师进行直接而尊重的沟通——安排一次正式会面，说明我的担忧并提出建设性的解决方案。我会准备一份书面研究进展报告，使谈话有专业框架。若直接沟通失败，我会向论文委员会成员、院系申诉专员或研究生院学生事务办公室寻求建议。在整个过程中，我会保留沟通和决策记录。我也会提醒自己：“办法总比问题多。”

Bioethics Knowledge Points

Mentor's responsibilities: Being available, allowing personality differences, letting trainees make their own decisions, teaching by words and examples, and continuing to

learn effective mentoring.

导师的责任：保持可接近，包容人格差异，允许受训者作出自己的决定，以言传身教进行指导，并持续学习有效的指导方式。

Mentee's responsibilities: Keeping regular contact, assessing strengths and goals, developing plans, following through on commitments, respecting the mentor's time, maintaining confidentiality, and being receptive to feedback.

学生的责任：保持定期联系，评估自身优势和目标，制定计划，履行承诺，尊重导师时间，维护保密，并愿意接受反馈。

Proactive engagement: The mentee should proactively write formal research progress reports at least annually, regardless of whether asked.

主动参与：无论是否被要求，学生都应至少每年主动撰写正式的研究进展报告。

Institutional oversight: Committee meetings, departmental monitoring, and university-level support systems exist as safety nets.

机构监督：委员会会议、院系监督和大学层面的支持系统都是安全网。

Case 3

Scenario: In the Smith lab, graduate student Chris is studying cancer cell metastasis. Senior postdoc Lee, seeking a faculty position, proposes a molecular cloning approach. A new postdoc, Alex, who has extensive cDNA cloning experience, provides Chris with crucial technical advice and hands-on assistance. Chris succeeds in identifying a key cDNA. Prof. Smith decides to submit a manuscript listing only Chris as co-author, excluding both Lee (who proposed the approach) and Alex (who provided essential technical help).

情景：在 Smith 实验室，研究生 Chris 正在研究癌细胞转移。正在寻求教职的高级博士后 Lee 提出了一种分子克隆方法。新博士后 Alex 具有丰富的 cDNA 克隆经验，为 Chris 提供了关键的技术建议和实际操作帮助。Chris 成功鉴定出一个关键 cDNA。Smith 教授决定投稿一篇论文，只将 Chris 列为共同作者，而排除提出方法的 Lee 和提供重要技术帮助的 Alex。

Analysis

Alex provided significant intellectual and technical contributions (experimental design, troubleshooting) that go beyond routine technical assistance.

Alex 提供了重要的智力和技术贡献（实验设计、故障排查），已经超出常规技术协助的范围。

Lee's conceptual contribution (proposing the molecular approach) also constitutes an intellectual contribution, even though he did not perform experiments.

Lee 的概念性贡献（提出分子方法）同样构成智力贡献，尽管他没有亲自做实验。

Excluding meaningful contributors from authorship raises questions about fairness and the proper allocation of credit.

将有实质贡献的人排除在作者名单之外，会引发公平性和合理分配学术信用的问题。

Prof. Smith, as PI, has significant power in determining authorship, but this power must be exercised ethically.

Smith 教授作为 PI 在决定作者身份方面拥有很大权力，但这种权力必须以伦理方式行使。

The case illustrates the tension between intellectual contribution, technical contribution, and the power dynamics in a lab hierarchy.

该案例体现了智力贡献、技术贡献以及实验室层级权力动态之间的张力。

What I Would Do

If I were Alex, I would first have a private, calm conversation with Chris and then with Prof. Smith, presenting a clear record of my contributions. I would reference my lab's or journal's authorship guidelines (e.g., ICMJE criteria) to support my case. If Prof. Smith was unresponsive, I would consider involving the departmental chair or an ombudsperson. I would not "just accept the situation" —proper authorship is both an honor and a right for those who contribute substantively. However, I would also avoid being confrontational,

focusing instead on presenting evidence-based arguments.

如果我是 Alex，我会先与 Chris 私下、冷静地沟通，然后再与 Smith 教授沟通，并清楚呈现我的贡献记录。我会引用实验室或期刊的作者署名指南（如 ICMJE 标准）来支持我的主张。如果 Smith 教授没有回应，我会考虑请院系主任或申诉专员介入。我不会“就这样接受”——恰当的作者身份既是荣誉，也是对实质贡献者的权利。不过，我也会避免对抗，把重点放在基于证据的论证上。

Bioethics Knowledge Points

Authorship criteria: Authorship means both honor and responsibility. Each author should have made an appropriate intellectual or practical contribution. Co-authorship implies endorsement of the work's integrity.

作者资格标准：作者身份意味着荣誉与责任并存。每位作者都应作出适当的智力或实践贡献。共同署名意味着认可该工作的完整性。

ICMJE standards: Substantial contributions to conception/design, data acquisition/analysis, manuscript drafting, and final approval are all authorship criteria.

ICMJE 标准：对构思/设计、数据获取/分析、手稿撰写以及最终批准作出实质贡献，都是作者资格的标准。

Acknowledgements vs. authorship: Minor contributions (e.g., routine technical support) may be acknowledged rather than granted authorship, but significant intellectual and technical contributions deserve authorship.

致谢与作者身份：较小贡献（如常规技术支持）可在致谢中列出而非授予作者身份，但重要的智力和技术贡献应获得作者身份。

Power dynamics in laboratories: PIs hold disproportionate power; ethical authorship practices require transparency and fairness regardless of hierarchy.

实验室中的权力动态：PI 拥有不成比例的权力；符合伦理的署名实践要求透明和公平，而不应受层级影响。

Case 4

Scenario: After a four-year study on giraffe growth hormones, a manuscript is being prepared. The contributors include: Billy Bob (undergraduate, provided the central model idea during a lab meeting), Jane (former postdoc, developed the key method, writing 20%), Peter (4th-year PhD student, designed most experiments, generated bulk data, writing 20%), Mary (undergraduate senior, contributed 30% data under Peter's guidance), Dick (new postdoc, did statistical analysis and optimized the method, writing 30%), Dr. Do-little (PI, provided funding/space but had minimal scientific input), and Dr. Seuss (rival PI whose overheard idea inspired the entire study).

情景：一项为期四年的长颈鹿生长激素研究正在准备投稿。贡献者包括：Billy Bob（本科生，在组会上提出核心模型想法）、Jane（前博士后，开发关键方法，撰写 20%）、Peter（四年级博士生，设计大部分实验、生成主要数据，撰写 20%）、Mary（高年级本科生，在 Peter 指导下贡献 30% 数据）、Dick（新博士后，做统计分析并优化方法，撰写 30%）、Do-little 博士（PI，提供经费/空间但几乎没有科学投入），以及 Seuss 博士（竞争对手 PI，其被无意听到的想法启发了整个研究）。

Analysis

Billy Bob's spontaneous contribution of the central model is an intellectual contribution, but his overall role (washing glassware) is limited. This raises a nuanced question: does a single key insight warrant authorship?

Billy Bob 自发提出核心模型是一项智力贡献，但他的整体角色（清洗玻璃器皿）较为有限。这提出了一个细微问题：单个关键洞见是否足以获得作者身份？

Jane's method development is a substantial contribution even though she left the lab —her departure does not erase her contribution.

Jane 的方法开发是实质性贡献，即使她已经离开实验室，她的离开也不会抹去其贡献。

Peter, as the primary experimenter and manuscript writer, has a strong claim to first authorship.

Peter 作为主要实验执行者和手稿撰写者，对第一作者身份有很强主张。

Dr. Do-little's lack of scientific input challenges the common practice of automatic PI authorship. Providing only funding and space may not meet authorship criteria.

Do-little 博士缺乏科学投入，这挑战了 PI 自动署名的常见做法。仅提供经费和空间可能并不满足作者资格标准。

Dr. Seuss was never formally involved, and her idea was obtained through eavesdropping, not collaboration —she cannot ethically be listed as an author but perhaps acknowledged.

Seuss 博士从未正式参与，该想法也是通过偷听而非合作获得——从伦理上说她不能被列为作者，但也许可以致谢。

What I Would Do

I would advocate for authorship based on substantive contribution: Peter as first author (primary experimentalist and writer), followed by Dick, Jane, and Mary as co-authors in order of relative contribution. Billy Bob would be acknowledged for his key conceptual insight but might not meet full authorship criteria unless further contributions were documented. Dr. Do-little would be corresponding/last author only if they provided genuine intellectual oversight (not just funding). Dr. Seuss would be acknowledged, not listed as co-author, since the idea was not shared in a collaborative context. I would push for an open lab discussion about authorship criteria before submission.

我会主张根据实质贡献来确定作者身份：Peter 作为第一作者（主要实验者和撰写者），随后按相对贡献将 Dick、Jane 和 Mary 列为共同作者。Billy Bob 会因其关键概念洞见获得致谢，但除非记录显示他还有进一步贡献，否则可能不满足完整作者资格。Do-little 博士只有在提供真正的智力指导（而不只是经费）时，才应作为通讯/末位作者。Seuss 博士应被致谢，而非列为共同作者，因为该想法并非在合作语境中分享。我会推动在投稿前进行一次公开的实验室署名标准讨论。

Bioethics Knowledge Points

Allocation of credit: Authorship, acknowledgements, and references/citations are the three formal ways to credit contributions.

信用分配：作者身份、致谢以及参考文献/引用，是正式给予贡献者信用的三种方式。

Gift/honorary authorship: Listing someone as an author solely because they are the PI, provided funding, or hold seniority —without substantive intellectual contribution —is considered gift authorship and is unethical.

赠予/荣誉作者身份：仅因某人是 PI、提供经费或资历较高，就在没有实质智力贡献的情况下将其列为作者，属于赠予作者身份，是不符合伦理的。

By signing as co-author, you state that you believe the manuscript represents honest work. This is both a privilege and a legal/ethical commitment.

签署为共同作者，即表示你相信手稿代表诚实的工作。这既是一项特权，也是一项法律和伦理承诺。

Discuss authorship early: Authorship order and criteria should be established at the beginning of a project, not after the manuscript is written.

尽早讨论作者身份：作者顺序和标准应在项目开始时确定，而不是等到手稿写完之后。

Case 5

Scenario: The U.S. Public Health Service enrolled nearly 400 African-American men with syphilis (and a control group without) in Tuskegee, Alabama. Although penicillin became a proven treatment around 1945, participants were deliberately left untreated so investigators could observe the natural progression of the disease. The study continued for 40 years until public exposure in 1972. Many subjects died, infected their wives, and had children born with congenital syphilis.

情景：美国公共卫生署在阿拉巴马州塔斯基吉招募了近 400 名患有梅毒的非裔美国男性（以及一个未患病对照组）。尽管青霉素在约 1945 年已成为被证实有效的治疗方法，参与者仍被故意置于不治疗状态，以便研究者观察疾病的自然进程。该研究持续了 40 年，直到 1972 年被公众曝光。许多受试者死亡，感染了妻子，并生下患有先天梅毒的孩子。

Analysis

The study grossly violated the principles of informed consent: participants were not told the true nature of the study and were actively deceived into believing they were receiving treatment.

该研究严重违反知情同意原则：参与者未被告知研究的真实性质，还被主动欺骗，以为自己正在接受治疗。

It exploited a vulnerable population (economically disadvantaged African-American men in the Jim Crow South) who had limited access to healthcare and information.

它剥削了脆弱群体（Jim Crow 制度下美国南方经济困难的非裔男性），他们获得医疗和信息的机会有限。

Even after an effective treatment was available, researchers withheld it —directly causing preventable suffering and death.

即使有效治疗已经存在，研究者仍然不给予治疗——这直接造成了本可避免的痛苦和死亡。

The study lacked any form of ethical oversight or review board approval.

该研究缺乏任何形式的伦理监督或审查委员会批准。

It fundamentally violated the principles of beneficence, non-maleficence, justice, and respect for autonomy.

它从根本上违反了行善、不伤害、公正以及尊重自主性的原则。

What I Would Do

This case requires no personal "if I were in this situation" analysis —it is unambiguously wrong. As a modern researcher, I would ensure that every study I design includes: rigorous IRB review, genuine informed consent, respect for vulnerable populations, and immediate provision of effective treatments if they become available. If I ever witnessed a study withholding effective treatment from participants, I would report it to the IRB, the institution, and if necessary, to regulatory authorities and the public. The Tuskegee case is a permanent

reminder that "scientific interest" can never justify harming human subjects.

这个案例无需进行个人“如果我处在这种情境中”的分析——它毫无疑问是错误的。作为现代研究者，我会确保自己设计的每一项研究都包括：严格的 IRB 审查、真正的知情同意、对脆弱群体的尊重，以及一旦有效治疗出现就立即提供治疗。如果我见到任何研究向参与者隐瞒有效治疗，我会向 IRB、所在机构，并在必要时向监管部门和公众报告。塔斯基吉案例永久提醒我们，“科学兴趣”绝不能成为伤害人体受试者的理由。

Bioethics Knowledge Points

Historical atrocities that shaped modern regulation: Tuskegee directly led to the Belmont Report (1979) and the establishment of IRB systems in the United States.

塑造现代监管的历史暴行：塔斯基吉事件直接促成了《贝尔蒙报告》（1979）以及美国 IRB 体系的建立。

Nuremberg Code (1947): Established voluntary consent as essential; arose from Nazi medical experiments.

《纽伦堡法典》（1947）：确立自愿同意为核心要求；其源于纳粹医学实验。

Declaration of Helsinki (1964, revised multiple times): The WMA's statement on ethical principles for medical research involving human subjects, emphasizing informed consent, privacy, protection of vulnerable populations, and the duty of physicians.

《赫尔辛基宣言》（1964，多次修订）：世界医学学会关于涉及人体受试者医学研究伦理原则的声明，强调知情同意、隐私、脆弱群体保护以及医生的义务。

Belmont Report principles: Respect for persons (autonomy & informed consent), Beneficence (maximize benefits, minimize harms), Justice (fair distribution of research burdens and benefits).

《贝尔蒙报告》原则：尊重人（自主性与知情同意）、行善（最大化收益、最小化伤害）、公正（公平分配研究负担和收益）。

IRB (Institutional Review Board): ≥ 5 members, not all from the same profession, at least 1 non-affiliated member. Responsible for reviewing and approving all human subjects

research.

IRB (机构审查委员会): 至少 5 名成员, 不应全部来自同一专业, 至少 1 名非本机构成员。负责审查并批准所有人体受试者研究。

Case 6

Scenario: Jim, a genetics graduate student, is working on a multi-center obesity genetics study. Samples are anonymized; a consent form states they will be used exclusively for this study. Renee, another graduate student, asks Jim to give her aliquots of his samples for her sickle-cell anemia research. She argues she only needs controls, is not doing a "study" on these people, and does not need much DNA. Jim stops her at the freezer and explains that she would need to either get re-consent from the obesity-study participants or order anonymous commercial samples.

情景: 遗传学研究生 Jim 正在参与一项多中心肥胖遗传学研究。样本已经匿名化; 同意书声明样本将专门用于该研究。另一名研究生 Renee 要求 Jim 给她一些样本等分液, 用于她的镰状细胞贫血研究。她认为自己只需要对照样本, 并不是在对这些人做“研究”, 而且不需要太多 DNA。Jim 在冰柜前阻止了她, 并解释说, 她要么需要让肥胖研究参与者重新同意, 要么订购匿名商业样本。

Analysis

The original informed consent explicitly stated that samples would be used exclusively for the obesity study. Using them for a different study without re-consent violates the consent agreement.

原始知情同意明确说明样本将专门用于肥胖研究。未经重新同意而将其用于另一项研究, 违反了同意协议。

Renee's argument that she is "not really doing a study" is irrelevant —any use outside the consented scope requires new ethical review and consent.

Renee 声称自己“并不是真的在做研究”并不相关——任何超出同意范围的使用都需要新的伦理审查和同意。

Jim acted correctly by stopping Renee and explaining the proper procedures.

Jim 阻止 Renee 并解释正确程序，是正确的做法。

Jim's suggestion to order anonymous commercial samples represents a practical, ethical alternative.

Jim 建议订购匿名商业样本，是一种实际可行且合乎伦理的替代方案。

Even well-intentioned shortcuts can lead to serious ethical and legal violations.

即使出于好意的捷径，也可能导致严重的伦理和法律违规。

What I Would Do

I would do exactly what Jim did: refuse to share the samples for a purpose outside the consented scope, explain why, and suggest ethical alternatives. If Renee insisted, I would escalate the matter to the study coordinator (Jane) or the PI. I would never authorize or facilitate the use of consented samples for an unauthorized purpose, no matter how minor the request seems. In my own research involving human-derived microbiome or clinical samples, I would always verify that my intended use falls within the scope of the original consent.

我会像 Jim 一样处理：拒绝为同意范围之外的目的分享样本，解释原因，并提出合乎伦理的替代方案。如果 Renee 坚持，我会将此事上报给研究协调员 Jane 或 PI。无论请求看似多么微小，我都不会授权或协助将已同意样本用于未经授权的目的。在我自己涉及人体来源微生物组或临床样本的研究中，我会始终核实拟定用途是否落在原始同意范围内。

Bioethics Knowledge Points

Informed consent is purpose-specific: Consent forms define the scope of permissible use. Any use outside that scope requires new consent or IRB waiver.

知情同意具有目的特异性：同意书界定了可允许使用的范围。任何超出该范围的使用都需要新的同意或 IRB 豁免。

The 8 elements of informed consent: Purpose and procedures, risks, benefits, alternatives,

confidentiality mechanisms, compensation for injury, contact information, and the right to withdraw.

知情同意的 8 个要素：目的和程序、风险、收益、替代方案、保密机制、受伤补偿、联系方式以及退出权。

Researcher independence on consent exemption: A researcher cannot independently determine that they are exempt from obtaining informed consent —only the IRB decides.

研究者不能自行决定同意豁免：研究者不能独立认定自己可免于获得知情同意——只有 IRB 可以作出决定。

Sample anonymization does not eliminate ethical obligations: Even anonymized samples are governed by the consent under which they were collected.

样本匿名化并不消除伦理义务：即使是匿名化样本，也仍受其收集时同意范围的约束。

Practical alternatives exist: Commercial DNA banks, de-identified public datasets, and biobanks with broad consent can often serve the same scientific purpose ethically.

实际替代方案存在：商业 DNA 库、去标识化公共数据集以及采用广泛同意的生物样本库，往往可以以合乎伦理的方式服务于相同科学目的。

Case 7

Scenario: A graduate student's advisor proposes a new series of experiments involving surgical manipulations in mice. The student realizes that the current IACUC-approved protocol does not include surgery. The student asks about submitting an amended protocol, but the advisor says he does not want to go through the trouble and suggests doing the experiments first —if the procedure looks promising, they can submit the amendment later.

情景：一名研究生的导师提出开展一系列新的小鼠外科操作实验。学生意识到当前 IACUC 批准的方案并不包括外科手术。学生询问是否应提交修订方案，但导师说他不费这个事，并建议先做实验——如果操作看起来有前景，再提交修订。

Analysis

The advisor's suggestion directly violates IACUC regulations. All animal experiments must be covered by an approved protocol before they begin.

导师的建议直接违反 IACUC 规定。所有动物实验在开始前都必须纳入已批准的方案。

The IACUC exists to protect animal welfare and ensure compliance with the 3Rs (Replace, Reduce, Refine).

IACUC 的存在是为了保护动物福利，并确保遵守 3R 原则（替代、减少、优化）。

The advisor's reasoning ("let's see if it works first") prioritizes convenience over regulatory compliance and animal welfare.

导师“先看看是否有效”的理由，把便利置于法规合规和动物福利之上。

As a graduate student, the power asymmetry makes it difficult to refuse, but compliance with regulations is non-negotiable.

作为研究生，权力不对等会使拒绝变得困难，但遵守规定是不可协商的。

If the unapproved experiments are discovered, both the student and the advisor face serious consequences (protocol suspension, loss of funding, institutional sanctions).

如果未经批准的实验被发现，学生和导师都将面临严重后果（方案暂停、经费损失、机构处分）。

What I Would Do

I would respectfully but firmly explain to my advisor that I cannot perform the experiments without an amended, IACUC-approved protocol. I would offer to prepare the amendment paperwork myself to reduce the administrative burden on my advisor. If my advisor insisted on proceeding without approval, I would seek guidance from another trusted faculty member, the IACUC office, or the departmental chair. I would not begin any unapproved procedures —the risk to the animals, my own career, and the lab's reputation is too great.

我会尊重但坚定地导师说明，没有经过修订并获得 IACUC 批准的方案，我不能

实施这些实验。我会主动提出由我准备修订材料，以减轻导师的行政负担。如果导师坚持未经批准就推进，我会向另一位可信任的教师、IACUC 办公室或院系主任寻求指导。我不会开始任何未经批准的程序——这对动物、我自己的职业生涯以及实验室声誉来说风险都太大。

Bioethics Knowledge Points

The 3Rs Principle (Russell & Burch, 1959): Replace (use alternatives to animals), Reduce (minimize the number of animals), Refine (minimize pain and suffering).

3R 原则 (Russell 与 Burch, 1959): 替代（使用动物替代方法）、减少（尽量减少动物数量）、优化（尽量减少疼痛和痛苦）。

IACUC (Institutional Animal Care and Use Committee): Requires at least 1 veterinarian (DVM), 1 non-affiliated member, and others. All animal protocols must be reviewed and approved before experiments begin.

IACUC (机构动物护理与使用委员会): 要求至少 1 名兽医 (DVM)、1 名非本机构成员以及其他成员。所有动物实验方案必须在实验开始前经过审查和批准。

Protocol amendments: Any change in experimental procedures (especially invasive ones like surgery) requires a formal protocol amendment and IACUC re-approval.

方案修订: 实验程序的任何改变（尤其是外科手术等侵入性操作）都需要正式方案修订并重新获得 IACUC 批准。

Power dynamics and ethical responsibility: Students must not compromise ethical standards due to pressure from advisors. Institutional channels exist for reporting and resolution.

权力动态与伦理责任: 学生不得因导师压力而牺牲伦理标准。机构渠道可用于报告和解决问题。

The biggest difference between human and animal subject regulations: In human subjects, the subject can refuse/withdraw. In animal subjects, the animal cannot advocate for itself, making institutional oversight even more critical.

人体与动物受试者监管的最大区别: 在人体研究中，受试者可以拒绝/退出；在动物

研究中，动物无法为自己发声，因此机构监督更加关键。

Case 8

Scenario: Dr. Leonard reviews a manuscript by Dr. Morris about a survivin-GFX protein interaction. He gives the paper to his graduate student Ms. Zane (without editor approval) for a detailed critique. Ms. Zane identifies a useful finding (survivin toxicity at high doses) that helps solve a problem in her own research. Dr. Leonard then uses the idea to develop a new experimental approach (stimulating endogenous GFX). Their resulting paper is published in Science. Dr. Morris later suspects that Dr. Leonard appropriated ideas from the manuscript under review.

情景: Leonard 博士审阅 Morris 博士关于 survivin-GFX 蛋白相互作用的一篇手稿。他在未获得编辑批准的情况下，将论文交给自己的研究生 Zane 女士作详细评论。Zane 女士发现了一个有用结果（高剂量 survivin 具有毒性），这帮助解决了她自己研究中的一个问题。随后 Leonard 博士利用这一想法发展出新的实验策略（刺激内源性 GFX）。他们由此产生的论文发表在 Science 上。后来 Morris 博士怀疑 Leonard 博士挪用了正在审稿手稿中的想法。

Analysis

Dr. Leonard violated peer review confidentiality by sharing the manuscript with Ms. Zane without the editor's permission. Peer review material is privileged information.

Leonard 博士未经编辑许可就与 Zane 女士分享手稿，违反了同行评审保密原则。同行评审材料属于特权信息。

Using ideas from a manuscript under review to advance one's own research is a serious breach of trust—it constitutes intellectual theft.

利用正在审稿手稿中的想法推进自己的研究，是严重的信任违背——构成智力盗用。

Even though Dr. Leonard used a "somewhat different" method, the conceptual foundation came from Dr. Morris's unpublished work.

尽管 Leonard 博士使用了“有些不同”的方法，但概念基础来自 Morris 博士尚未发表的工作。

Ms. Zane bears some responsibility —she benefited from privileged information, though she may not have fully understood the ethical implications.

Zane 女士也承担一定责任——她从特权信息中受益，尽管她可能并未完全理解其中的伦理含义。

Dr. Morris's suspicion is well-founded: Dr. Leonard had no prior work on GFX, making the connection to the reviewed manuscript obvious.

Morris 博士的怀疑是有根据的：Leonard 博士此前没有关于 GFX 的工作，这使其与
被审手稿之间的联系非常明显。

What I Would Do

If I were asked to review a manuscript in my field, I would treat it as strictly confidential. I would not share it with students or colleagues without explicit permission from the editor. If a student could help with the review (as an educational exercise), I would first seek the editor's approval. I would never use unpublished ideas from a manuscript under review to redirect my own research. If I recognized a conflict of interest (e.g., the manuscript relates closely to my ongoing work), I would decline the review or disclose the conflict to the editor.

如果我被要求审阅本领域的手稿，我会将其视为严格保密材料。未经编辑明确许可，我不会与学生或同事分享。如果学生可以协助评审（作为教育练习），我会先征得编辑批准。我绝不会利用正在审稿手稿中的未发表想法来调整自己的研究方向。如果我意识到存在利益冲突，我会拒绝审稿或向编辑披露冲突。

Bioethics Knowledge Points

Confidentiality of peer review: Material under review is privileged information. Reviewers must not share it, discuss it with unauthorized parties, or use it for personal gain.

同行评审的保密性：待审材料属于特权信息。审稿人不得分享、与未经授权者讨论，

或将其用于个人利益。

Responsible peer review: Includes timely review, honest evaluation, conflict-of-interest disclosure, and maintaining confidentiality.

负责任的同行评审：包括及时审稿、诚实评价、披露利益冲突以及维护保密性。

Challenges of anonymous peer review: Anonymous reviewers may abuse their position by holding up reviews, appropriating information, or giving unfair reviews —the very concerns this case illustrates.

匿名同行评审的挑战：匿名审稿人可能滥用地位，拖延评审、挪用信息或给出不公平评审——本案例正体现了这些担忧。

Recourse for plagiarized/stolen ideas: The original author can contact the alleged offender directly, seek institutional investigation, contact the journal editor, or pursue formal misconduct proceedings.

被剽窃/被盗想法的救济途径：原作者可以直接联系涉嫌违规者，寻求机构调查，联系期刊编辑，或启动正式不端行为程序。

Conflicts of interest in peer review: Financial interests, personal relationships, professional competition, and intellectual proximity can all bias reviews.

同行评审中的利益冲突：财务利益、个人关系、职业竞争和智力接近性都可能使评审产生偏差。

Case 9

Scenario: You are a junior researcher in Dr. Leyos's lab. You discover that a senior researcher —who is a close friend of Dr. Leyos and has co-published multiple papers —has been manipulating data to create statistical significance in some experiments. Dr. Leyos plans to co-author a paper with this person, and you suspect Dr. Leyos may not know about the manipulation.

情景：你是 Leyos 博士实验室的一名初级研究者。你发现一名资深研究者——他是 Leyos 博士的密友，并已共同发表多篇论文——一直在操纵数据，使某些实验产生

统计学显著性。Leyos 博士计划与此人共同署名一篇论文，而你怀疑 Leyos 博士可能并不知道数据被操纵。

Analysis

Deliberate data manipulation to achieve statistical significance is falsification —one of the three pillars of research misconduct (FFP).

为了获得统计学显著性而故意操纵数据，属于篡改——科研不端三大类型（FFP）之一。

This is not an honest error or negligence; it is intentional deception.

这不是诚实错误或疏忽，而是有意欺骗。

The junior researcher faces a classic whistleblower dilemma: the personal risk (retaliation, career damage) vs. the ethical obligation to report.

初级研究者面临典型的举报者困境：个人风险（报复、职业受损）与举报的伦理义务之间的冲突。

Dr. Leyos's close friendship with the offender creates a potential bias —the PI may be unwilling to believe or act on the report.

Leyos 博士与违规者关系密切，形成潜在偏见——PI 可能不愿相信或处理报告。

If the falsified data is published, it harms the entire scientific community by introducing unreliable findings into the literature.

如果伪造的数据发表，将把不可靠结果引入文献，损害整个科学共同体。

What I Would Do

Before taking action, I would carefully consider: (1) Am I certain about what I observed? (2) Do I have documented evidence? (3) What are the institutional channels for reporting? I would first try to verify my observations by rechecking the data independently. If confirmed, I would consider discussing my concerns with Dr. Leyos privately, presenting the evidence objectively. If Dr. Leyos dismisses my concerns, I would escalate to the departmental research integrity officer or the institutional Office of Research Integrity. I would

document everything carefully throughout the process. I would not confront the senior researcher directly, as this could lead to evidence destruction or retaliation.

在采取行动前，我会认真考虑：（1）我是否确定自己观察到的情况？（2）我是否有文档化证据？（3）机构内有哪些报告渠道？我会先通过独立复核数据来验证观察结果。若确认无误，我会考虑私下向 Leyos 博士说明担忧，并客观呈现证据。如果 Leyos 博士忽视我的担忧，我会向院系科研诚信负责人或机构科研诚信办公室上报。在整个过程中，我会仔细记录一切。我不会直接质问那名资深研究者，因为这可能导致证据被销毁或引发报复。

Bioethics Knowledge Points

Fabrication, Falsification, Plagiarism (FFP): The three defined categories of research misconduct. Falsification means manipulating materials, equipment, processes, or changing/omitting data such that research is not accurately represented.

编造、篡改、剽窃 (FFP): 科研不端的三类定义。篡改是指操纵材料、设备、过程，或改变/遗漏数据，使研究未被准确呈现。

Image manipulation standards: Any change that conceals information is unacceptable — changing brightness/contrast to hide signals, using clone tools to remove data, or selectively cropping images to remove unfavorable results.

图像处理标准: 任何掩盖信息的改变都是不可接受的——例如改变亮度/对比度以隐藏信号，使用克隆工具移除数据，或选择性裁剪图像以去除不利结果。

Consequences of misconduct: Damage to science and society, betrayal of colleagues and the scientific community, and ruination of one's own career and institution's reputation.

不端行为的后果: 损害科学与社会，背叛同事和科学共同体，并毁掉个人职业生涯和所在机构声誉。

Criteria for misconduct: Significant departure from accepted practices, committed intentionally/knowingly/recklessly, and proven by a preponderance of evidence.

不端行为判定标准: 显著偏离公认实践，有意/明知/鲁莽地实施，并由优势证据证明。

Whistleblower protections: Institutions should have policies protecting those who report misconduct in good faith.

举报者保护：机构应有政策保护善意报告不端行为的人。

Case 10

Scenario: Sandra is a graduate student in Dr. Frederick's lab, whose project is funded entirely by a single company. Although Dr. Frederick initially assured Sandra that the corporate funding would not conflict with her education, she gradually realizes that the company's research needs dictate her experiments, limiting her ability to pursue basic research questions, develop her own ideas, or build a coherent dissertation. She is also pressured to sign confidentiality statements that prevent her from discussing her work with peers, making it difficult to get advice.

情景：Sandra 是 Frederick 博士实验室的一名研究生，其项目完全由一家公司资助。尽管 Frederick 博士最初向 Sandra 保证企业经费不会与她的教育冲突，但她逐渐意识到，公司的研究需求支配了她的实验，限制了她追求基础研究问题、发展个人想法或形成连贯学位论文的能力。她还被要求签署保密声明，禁止与同伴讨论工作，使她难以获得建议。

Analysis

This is a classic conflict of commitment: the tension between academic training (which demands open inquiry, free publication, and intellectual development) and industry obligations (which demand focused deliverables, confidentiality, and commercial relevance).

这是典型的职责冲突：学术训练（要求开放探索、自由发表和智力发展）与产业义务（要求聚焦交付物、保密和商业相关性）之间的张力。

Dr. Frederick bears responsibility for assuring Sandra that the arrangement would not conflict with her education — this assurance proved inaccurate.

Frederick 博士曾向 Sandra 保证这种安排不会影响她的教育，因此对此负有责任——事实证明这一保证并不准确。

Sandra's inability to publish or discuss her work undermines the fundamental principles of academic training and scientific openness.

Sandra 无法发表或讨论自己的工作，破坏了学术训练和科学开放的基本原则。

The confidentiality constraints isolate Sandra from the intellectual community that is essential for her development.

保密限制使 Sandra 与其发展所必需的智力共同体相隔离。

The power relationship between Sandra and her advisor makes it particularly difficult for her to negotiate changes.

Sandra 与导师之间的权力关系，使她尤其难以协商改变。

What I Would Do

I would have a frank conversation with Dr. Frederick, presenting specific examples of how the industry constraints are limiting my academic development. I would propose solutions: dedicating a portion of my time to independent research, negotiating with the company for delayed publication rights, or identifying aspects of my work that can be published without violating confidentiality. If Dr. Frederick was unwilling or unable to help, I would consult the thesis committee, the graduate school, or consider transitioning to a different project. Before joining any industry-funded project in the future, I would establish clear written agreements about publication rights, intellectual freedom, and the scope of confidentiality obligations.

我会与 Frederick 博士坦诚交谈，提出产业限制如何影响我学术发展的具体例子。我会提出解决方案：将一部分时间用于独立研究，与公司协商延迟发表权，或确定可以在不违反保密要求的情况下发表的工作内容。如果 Frederick 博士不愿或不能帮助，我会咨询论文委员会、研究生院，或考虑转到其他项目。今后在加入任何产业资助项目前，我会就发表权、学术自由和保密义务范围建立清晰的书面协议。

Bioethics Knowledge Points

Conflict of commitment (academia vs. industry): Arises when industry funding restricts what questions can be asked, what results can be published, and how students are trained.

职责冲突 (学术界 vs. 产业界): 当产业经费限制可以提出哪些问题、哪些结果可以发表以及学生如何接受训练时, 就会产生职责冲突。

Ethical problems in science are not always obvious: Sometimes they arise gradually from pressure, dependence, and power relationships rather than from a single dramatic event.

科学中的伦理问题并不总是显而易见: 有时问题不是源于单一戏剧性事件, 而是逐渐来自压力、依赖和权力关系。

Industry-university collaboration: Can be enormously productive but requires clear agreements on publication rights, IP ownership, student training, and academic freedom.

产学合作: 可以极具成效, 但需要就发表权、知识产权归属、学生培养和学术自由达成清晰协议。

The factor that most limits Sandra is her dependence on her advisor —highlighting how power asymmetries can perpetuate ethical problems.

最限制 Sandra 的因素是她对导师的依赖——这凸显了权力不对等如何延续伦理问题。

Bayh-Dole Act (1980): Permits universities to patent federally funded discoveries and transfer them to the private sector, but also requires disclosure and commercialization.

Bayh-Dole 法案 (1980): 允许大学将联邦资助产生的发现申请专利并转移给私营部门, 但也要求披露和商业化。

Case 11

Scenario: Mr. Asset, a graduate student working for Dr. Bond, conducts proprietary polymer research funded by Chemical Industries, Inc. The company's technical liaison, Mr. Cash, enthusiastically shares that the recent results are the final data needed to market a

breakthrough product, and that the company's stock would soar upon announcement. That evening, Mr. Asset tells his wife and brother-in-law (an investment banker) about Mr. Cash's excitement and expectations. The brother-in-law then advises his clients to buy Chemical Industries' stock.

情景：Asset 先生是 Bond 博士指导下的一名研究生，正在开展由 Chemical Industries, Inc. 资助的专有聚合物研究。公司的技术联络人 Cash 先生兴奋地表示，最近的结果正是将一项突破性产品推向市场所需的最终数据，并称公司股票会在公告后大涨。当晚，Asset 先生把 Cash 先生的兴奋和预期告诉了妻子和身为投资银行家的内兄。随后内兄建议其客户购买 Chemical Industries 的股票。

Analysis

Mr. Asset violated the confidentiality agreement of the industry-sponsored research by disclosing proprietary results to family members.

Asset 先生向家庭成员披露专有结果，违反了产业资助研究的保密协议。

Even though Mr. Asset may not have intended financial manipulation, his disclosure directly led to insider trading by his brother-in-law.

即使 Asset 先生可能无意进行财务操纵，他的披露也直接导致内兄进行内幕交易。

The chain of responsibility extends from Mr. Asset (breach of confidentiality) to the brother-in-law (insider trading) —both committed ethical and potentially legal violations.

责任链从 Asset 先生（违反保密）延伸到内兄（内幕交易）——两者都犯下伦理上并可能在法律上的违规行为。

Mr. Cash may also bear some responsibility for sharing commercially sensitive information so enthusiastically, though he likely expected confidentiality from Mr. Asset.

Cash 先生如此兴奋地分享商业敏感信息，也可能承担一定责任，尽管他很可能期待 Asset 先生保密。

This case shows that confidentiality in industry-sponsored research extends beyond the lab—it includes all conversations, even casual dinner talk.

该案例表明，产业资助研究中的保密义务超出实验室范围——它包括所有谈话，甚

至是随意的晚餐闲聊。

What I Would Do

I would never discuss proprietary or confidential research results with anyone outside the authorized team—including family members. If excited about a breakthrough, I would channel that enthusiasm into writing a progress report for my PI and the company, rather than sharing it informally. I would treat all confidentiality agreements with the same seriousness as IRB protocols. If I were unsure about what I could or could not discuss, I would consult Dr. Bond or the company's legal representative before sharing anything.

我绝不会与授权团队之外的任何人讨论专有或保密研究结果——包括家庭成员。如果对突破感到兴奋，我会把这种热情转化为给 PI 和公司的进展报告，而不是非正式分享。我会像对待 IRB 方案一样严肃对待所有保密协议。如果不确定哪些内容可以或不可以讨论，我会在分享任何信息前咨询 Bond 博士或公司的法律代表。

Bioethics Knowledge Points

Confidentiality in industry-sponsored research: Researchers must strictly adhere to confidentiality agreements—breaches can have legal, financial, and career consequences.

产业资助研究中的保密性：研究者必须严格遵守保密协议——违约可能带来法律、财务和职业后果。

Insider trading: Using non-public, material information to trade securities is illegal and constitutes a serious violation of trust.

内幕交易：利用非公开、重大信息交易证券是非法的，也构成严重的信任违背。

Sharing does not alter ownership: Sharing research materials or information (with proper agreements like MTAs) does not change who owns the IP, but unauthorized sharing breaches the agreement entirely.

共享并不改变所有权：共享研究材料或信息（在 MTA 等适当协议下）并不改变知识产权归属，但未经授权的共享会完全违反协议。

Material Transfer Agreement (MTA): A formal agreement required whenever research

materials are shared between institutions —published materials should be available to academic researchers, but with proper documentation.

材料转移协议 (MTA)：研究材料在机构之间共享时所需的正式协议——已发表材料应向学术研究者开放，但必须有适当文件记录。

The boundary between professional and personal life: Ethical obligations do not stop at the lab door.

职业生活与私人生活的边界：伦理义务不会止于实验室门口。

Case 12

Scenario: Dr. Alice Charles, a mid-career scientist, discovers that Dr. Chris Long, a department chair, has incorporated large portions of her previous publication into his own chapter without attribution. When confronted, Dr. Long denies it, then admits the language is "remarkably similar," blaming his junior lab members who wrote portions of the chapter under time pressure. He argues that the original research was based on his lab's work anyway and admits only to negligence in supervising his subordinates.

情景：中年科学家 Alice Charles 博士发现，系主任 Chris Long 博士在自己的章节中大量使用了她先前发表论文的内容，却没有注明出处。被质询时，Long 博士先是否认，随后承认文字“非常相似”，并把责任归咎于在时间压力下撰写部分章节的初级实验室成员。他辩称原始研究本来就是基于其实验室工作，并且只承认自己在监督下属方面存在疏忽。

Analysis

Dr. Long's act is plagiarism regardless of whether he personally copied the text or delegated the writing. As the listed author, he is responsible for the integrity of the entire work.

无论 Long 博士是亲自复制文字，还是委托他人写作，他的行为都是剽窃。作为署名作者，他对整个作品的完整性负责。

His defense that "the original research came from his lab" is irrelevant —citing the source is

required even when building on one's own previous work (self-plagiarism is also a concern).

他辩称“原始研究来自自己的实验室”并不相关——即使是在自己先前工作的基础上推进，也必须引用来源（自我剽窃同样值得关注）。

Blaming junior members while taking credit as senior author demonstrates a failure of supervisory responsibility.

一边作为高级作者获取信用，一边将责任推给初级成员，体现了监督责任的失败。

The subordinates who actually wrote the text were not acknowledged either, compounding the ethical failure.

实际撰写文字的下属也没有得到致谢，这进一步加重了伦理问题。

Dr. Charles is justified in reporting the matter to Dr. Long's dean and the editor.

Charles 博士向 Long 博士的院长和编辑报告此事是有正当理由的。

What I Would Do

If I were Dr. Charles, I would do exactly as she did: confront Dr. Long with evidence, demand a formal acknowledgment and correction, and if unsatisfied with his response, report the matter to his institution's dean and the journal editor. If I were a junior researcher in Dr. Long's lab and realized I had been asked to write text that turned out to be plagiarized from another source, I would report the situation to the department's research integrity officer. As a PI myself, I would never delegate writing without providing the source materials and requiring proper citation —and I would review all text written by my team members before submission.

如果我是 Charles 博士，我会像她一样处理：拿出证据质询 Long 博士，要求正式致谢和更正；如果对其回应不满意，就向其所在机构院长和期刊编辑报告。如果我是 Long 博士实验室的初级研究者，并意识到自己被要求撰写的文字其实剽窃自其他来源，我会向院系科研诚信负责人报告。作为 PI，我绝不会在不提供来源材料并要求规范引用的情况下委托写作——并且会在投稿前审阅团队成员撰写的全部文字。

Bioethics Knowledge Points

Author responsibility: By signing as a co-author (or sole author), you are stating that you believe the manuscript represents honest work. Authors are responsible for the integrity of everything in the paper.

作者责任: 签署为共同作者（或唯一作者）时，你是在声明自己相信手稿代表诚实的工作。作者对论文中所有内容的完整性负责。

Supervisory responsibility: PIs and senior authors cannot escape responsibility by blaming junior team members. The obligation to ensure proper citation and originality rests with the listed authors.

监督责任: PI 和高级作者不能通过责怪初级团队成员来逃避责任。确保规范引用和原创性的义务由署名作者承担。

Handling misconduct: Institutional channels (deans, research integrity officers, journal editors) exist for reporting and investigating misconduct.

处理不端行为: 院长、科研诚信负责人、期刊编辑等机构渠道可用于报告和调查不端行为。

Self-plagiarism: Even recycling one's own previously published text without proper citation is considered a form of plagiarism.

自我剽窃: 即使重复使用自己先前已发表文字而不作适当引用，也被视为一种剽窃。

Retraction and correction policies: Journals distinguish between Erratum (publisher error), Corrigendum (author error), and Retraction (severe misconduct including plagiarism, data fabrication, or fraud).

撤稿与更正政策: 期刊区分 Erratum（出版方错误）、Corrigendum（作者错误）和 Retraction（包括剽窃、数据编造或欺诈在内的严重不端）。

Key Principles & Personal Reflections

Key Principles That Demand Special Attention

The FFP Triad Is the Absolute Boundary

Fabrication (making up data), Falsification (manipulating data so research is not accurately represented), and Plagiarism (using others' ideas/words without credit) constitute the three defined categories of research misconduct. Unlike honest errors or even negligence, FFP involves intentional or reckless deception. The consequences are severe and irreversible — retraction of papers, loss of funding, career destruction, and damage to the entire scientific community. As a researcher, I must internalize that **no result is worth fabricating, no curve is worth smoothing dishonestly, and no paragraph is worth copying without attribution.**

编造（捏造数据）、篡改（操纵数据导致研究未被准确呈现）和剽窃（使用他人的想法/文字而不署名）构成科研不端的三类定义。不同于诚实错误甚至疏忽，FFP 涉及有意或鲁莽的欺骗。其后果严重且难以逆转——论文撤回、经费丧失、职业毁灭，并损害整个科学共同体。作为研究者，我必须内化这一点：**没有任何结果值得编造，没有任何曲线值得不诚实地平滑，没有任何段落值得在不注明出处的情况下复制。**

Informed Consent Is Non-Negotiable and Purpose-Specific

Whether dealing with human subjects, human-derived samples, or even anonymized data, the principle of informed consent must be rigorously upheld. The Tuskegee tragedy and the subsequent Nuremberg Code, Declaration of Helsinki, and Belmont Report all remind us that science does not have the right to use people as mere instruments. Consent must be voluntary, informed, and specific to the study at hand. Researchers cannot independently determine exemption —only the IRB can make that judgment. In the era of big data and biobanks, this principle extends to digital consent, re-consent for secondary use, and the ethical handling of incidental findings.

无论面对人体受试者、人体来源样本，还是匿名化数据，都必须严格维护知情同意原则。塔斯基吉悲剧及其后的《纽伦堡法典》、《赫尔辛基宣言》和《贝尔蒙报告》都提醒我们，科学没有权利把人仅仅当作工具。同意必须是自愿的、知情的，并且

具体对应当前研究。研究者不能自行认定豁免——只有 IRB 可以作出判断。在大数据和生物样本库时代，这一原则延伸到数字同意、二次使用的重新同意以及偶然发现的伦理处理。

The 3Rs and IACUC Are Moral Obligations, Not Administrative Burdens

Replace, Reduce, Refine —these are not bureaucratic checkboxes but ethical commitments to the welfare of sentient beings. Animals cannot consent or withdraw. The IACUC protocol exists precisely because animals have no voice. Researchers must never begin unapproved experiments, and protocol amendments must be submitted before any procedural changes. The moral weight of using an animal in research demands that we justify every experiment, minimize every suffering, and explore every alternative.

替代、减少、优化——这些不是官僚式的勾选框，而是对有感知能力生命福祉的伦理承诺。动物不能同意或退出。IACUC 方案之所以存在，正是因为动物没有自己的声音。研究者绝不能开始未经批准的实验，任何程序改变前都必须提交方案修订。使用动物进行研究所承载的道德重量，要求我们为每个实验提供正当理由，尽量减少每一种痛苦，并探索每一种替代方案。

Authorship Is Both Honor and Responsibility

Authorship should reflect substantive intellectual contribution, not seniority, funding power, or social relationships. Gift authorship (adding a PI who did not contribute) and ghost authorship (excluding contributors) are both ethical violations. Authorship discussions should happen early in a project, be documented, and follow recognized standards (ICMJE, CRediT). By signing as a co-author, one affirms the integrity of the entire work —this is a weighty commitment.

作者身份应反映实质性智力贡献，而不是资历、经费权力或社会关系。赠予作者身份（加入没有贡献的 PI）和幽灵作者身份（排除贡献者）都是伦理违规。作者身份讨论应在项目早期进行、形成记录，并遵循公认标准（ICMJE、CRediT）。签署为共同作者，就是在确认整项工作的完整性——这是一项沉甸甸的承诺。

Peer Review Confidentiality Is Sacred

Peer review is the backbone of scientific quality control, and it depends entirely on trust. Reviewers must treat submitted manuscripts and grant proposals as privileged information. Sharing review materials without editor permission, appropriating ideas, or using review as a competitive tool all undermine the integrity of the system. The anonymity of reviewers is both a protection and a responsibility.

同行评审是科学质量控制的支柱，并且完全依赖信任。审稿人必须将投稿手稿和基金申请视为特权信息。未经编辑许可分享评审材料、挪用想法，或将评审作为竞争工具，都会破坏系统完整性。审稿人的匿名性既是一种保护，也是一种责任。

Conflicts of Interest Must Be Disclosed, Not Hidden

Financial interests, personal relationships, institutional affiliations, ideological commitments —any factor that could bias one's professional judgment must be disclosed. The problem is not having a conflict of interest (which is often unavoidable) but failing to disclose it. Transparency allows the community to evaluate findings in context. This extends to industry-sponsored research, where confidentiality constraints, publication restrictions, and commercial pressures can subtly distort the research process.

财务利益、个人关系、机构隶属、意识形态承诺——任何可能影响专业判断的因素都必须披露。问题不在于存在利益冲突（这往往难以避免），而在于隐瞒不报。透明性使共同体能够在语境中评价研究发现。这也延伸到产业资助研究，因为保密限制、发表限制和商业压力都可能微妙地扭曲研究过程。

Errors and Negligence Are Not Misconduct —But They Require Action

Honest errors (arising from technique limitations, human fallibility, or the state of knowledge) are unavoidable and demonstrate the self-correcting nature of science —provided they are acknowledged promptly and openly. Negligence (carelessness, haste, inattention) is more serious but still distinct from intentional misconduct. The key principle is: **acknowledge promptly, fix immediately, and be as accurate and careful as possible.** Every scientific result must be carefully verified, submitted to peer review, and scrutinized after publication.

诚实错误（源于技术限制、人类易错性或知识状态）不可避免，并且体现了科学的自我纠错性——前提是它们被及时、公开地承认。疏忽（粗心、仓促、注意力不足）更严重，但仍不同于故意不端。关键原则是：**及时承认，立即修正，并尽可能准确和谨慎**。每一项科学结果都必须被认真验证，提交同行评审，并在发表后继续接受审视。

The Mentor-Mentee Relationship Is an Ethical Structure, Not Just a Personal One

The relationship between advisor and student is one of the most consequential in a researcher's career. It is governed by mutual responsibilities, institutional expectations, and ethical norms. Mentors must be available, supportive, and respectful of their trainees' autonomy. Mentees must be proactive, communicative, and committed. When the relationship becomes dysfunctional, institutional channels (committees, ombudspersons, graduate schools) exist to provide support. The mentee must remember: **you are the first responsible person for your own career development.**

导师与学生之间的关系，是研究者职业生涯中最重要的关系之一。它受共同责任、机构期待和伦理规范约束。导师必须可接近、支持学生，并尊重受训者自主性。学生必须主动、善于沟通并保持投入。当关系出现功能失调时，机构渠道（委员会、申诉专员、研究生院）可以提供支持。学生必须记住：**你是自己职业发展的第一责任人。**

Personal Reflections

As a researcher working in microbiomics and virus-phageomics —fields that sit at the intersection of ecology, medicine, data science, and biotechnology —bioethics is not an abstract concern but a daily operational reality. Here are the principles I commit to upholding in my future research:

作为一名从事微生物组学与病毒—噬菌体组学研究的研究者——这些领域位于生态学、医学、数据科学和生物技术的交叉处——生命伦理并不是抽象问题，而是日常运行现实。以下是我承诺在未来研究中坚持的原则：

On Human-Derived Samples and Data

Microbiome research frequently involves human-derived samples (fecal, oral, skin, gut, etc.) that are deeply personal and sometimes stigmatized. I must ensure that all sample collection follows rigorous informed consent procedures, with participants fully aware of how their samples and associated metadata will be used, stored, and shared. In metagenomics and viromics, sequencing data can sometimes incidentally reveal host genetic information—I must have protocols for handling such incidental findings ethically. When depositing sequencing data to public databases (e.g., NCBI SRA, EBI ENA), I must ensure proper de-identification and compliance with the original consent scope.

微生物组研究经常涉及人体来源样本（粪便、口腔、皮肤、肠道等），这些样本具有高度个人性，有时还带有污名化风险。我必须确保所有样本采集都遵循严格的知情同意程序，让参与者充分了解其样本及相关元数据将如何被使用、保存和共享。在宏基因组学与病毒组学中，测序数据有时可能偶然揭示宿主遗传信息——我必须有伦理处理这类偶然发现的方案。当向公共数据库（如 NCBI SRA、EBI ENA）提交测序数据时，我必须确保适当去标识化，并符合原始同意范围。

On Data Integrity and Reproducibility

Microbiome and phageome data analysis involves complex bioinformatics pipelines with many subjective decisions (quality filtering thresholds, taxonomic databases, statistical methods). Each decision can dramatically affect results. I commit to:

微生物组和噬菌体组数据分析涉及复杂的生物信息学流程，其中包含许多主观决定（质量过滤阈值、分类数据库、统计方法）。每一个决定都可能显著影响结果。我承诺：

Documenting every analytical step with version-controlled code and parameter records.

用版本控制代码和参数记录来记录每一个分析步骤。

Avoiding cherry-picking results or selectively reporting only significant findings (p-hacking).

避免挑选结果或只选择性报告显著发现 (p-hacking)。

Sharing raw data, analysis scripts, and workflows to enable reproducibility.

共享原始数据、分析脚本和工作流程，以促进可重复性。

Clearly distinguishing between exploratory and confirmatory analyses.

清楚区分探索性分析与验证性分析。

Acknowledging the limitations of computational predictions (e.g., phage-host predictions, functional annotations) rather than over-interpreting them.

承认计算预测（如噬菌体—宿主预测、功能注释）的局限性，而不是过度解读。

On Collaboration and Authorship

Microbiomics is inherently interdisciplinary —collaborations with clinicians, bioinformaticians, ecologists, and chemists are the norm. I must:

微生物组学本质上是跨学科的——与临床医生、生物信息学家、生态学家和化学家的合作是常态。我必须：

Establish clear authorship agreements at the beginning of every collaboration.

在每次合作开始时建立清晰的作者身份协议。

Properly credit bioinformaticians and data analysts who contribute substantially to the intellectual content.

恰当认可对智力内容作出实质贡献的生物信息学家和数据分析人员。

Use MTAs when sharing microbial strains, phage isolates, or other biological materials.

共享微生物菌株、噬菌体分离株或其他生物材料时使用 MTA。

Respect the intellectual property of collaborators and avoid using shared unpublished data for purposes outside the agreed scope.

尊重合作者的知识产权，避免将共享的未发表数据用于约定范围之外的目的。

On Dual-Use Concerns and Biosafety

Phage research, while often celebrated for its therapeutic potential (phage therapy), also involves manipulating biological entities that can lyse bacteria. I must be mindful of:

噬菌体研究虽然常因其治疗潜力（噬菌体疗法）而受到赞誉，但也涉及操纵能够裂解细菌的生物实体。我必须注意：

The dual-use potential of engineered phages or phage-derived antimicrobials.

工程化噬菌体或噬菌体来源抗菌剂的双重用途潜力。

Biosafety regulations for working with novel phages, especially those isolated from clinical samples.

处理新型噬菌体的生物安全规定，尤其是从临床样本中分离出的噬菌体。

The responsible disclosure of research findings that could be misused (e.g., engineering phages against beneficial microorganisms).

对可能被滥用的研究发现进行负责任披露（例如工程化针对有益微生物的噬菌体）。

Environmental release considerations if phage-based biocontrol agents are developed.

如果开发噬菌体生物防治剂，需要考虑环境释放问题。

On Publication and Transparency

In the competitive landscape of microbiome and viromics research, I commit to:

在微生物组和病毒组研究竞争激烈的格局中，我承诺：

Publishing negative results and null findings, which are crucial for the field's maturation.

发表阴性结果和零结果，因为它们对该领域的成熟至关重要。

Disclosing all conflicts of interest, including any commercial interests in probiotic, phage therapy, or diagnostic applications.

披露所有利益冲突，包括在益生菌、噬菌体治疗或诊断应用中的任何商业利益。

Practicing responsible peer review —maintaining confidentiality, providing fair and constructive feedback, and declining reviews where I have conflicts.

负责任地进行同行评审——维护保密，提供公平而建设性的反馈，并在存在利益冲突时拒绝审稿。

Reporting all results accurately, even when they contradict popular hypotheses or my own previous findings.

准确报告所有结果，即使它们与流行假说或我自己此前的发现相矛盾。

Final Thought

Bioethics is not a set of rules imposed from outside —it is the conscience of the scientific community, internalized by each researcher. Throughout this course, I have come to understand that ethical violations rarely begin as dramatic acts of deception. More often, they start with small compromises: a shortcut here, a selective omission there, a moment of inattention or a reluctance to challenge authority. The most important lesson is vigilance —**maintaining awareness that every decision in the research process, no matter how small, carries ethical weight.**

生命伦理不是从外部强加的一套规则——它是科学共同体的良知，并由每位研究者内化。在本课程中，我逐渐认识到，伦理违规很少始于戏剧性的欺骗行为。更多时候，它们从微小妥协开始：这里走一次捷径，那里选择性省略一次，一次疏忽，或一次不愿挑战权威。最重要的教训是保持警觉——**意识到研究过程中每一个决定，无论多么微小，都承载伦理重量。**

As I pursue my career in microbiomics and virus-phageomics, I carry with me the lessons of Tuskegee, the principles of Helsinki, the 3Rs of Russell and Burch, and the wisdom of mentors who taught me that **the integrity of the process matters as much as the significance of the result.** Science is a privilege and a public trust. I intend to honor both.

在追求微生物组学与病毒—噬菌体组学职业道路时，我会带着塔斯基吉的教训、赫尔辛基原则、3R 原则，以及导师们传递给我的智慧：**过程的完整性与结果的重要性同样重要。**科学是一种特权也是一项公共信任。我愿同时珍惜并维护。